Studying Laser-Plasma Instabilities with EPOCH

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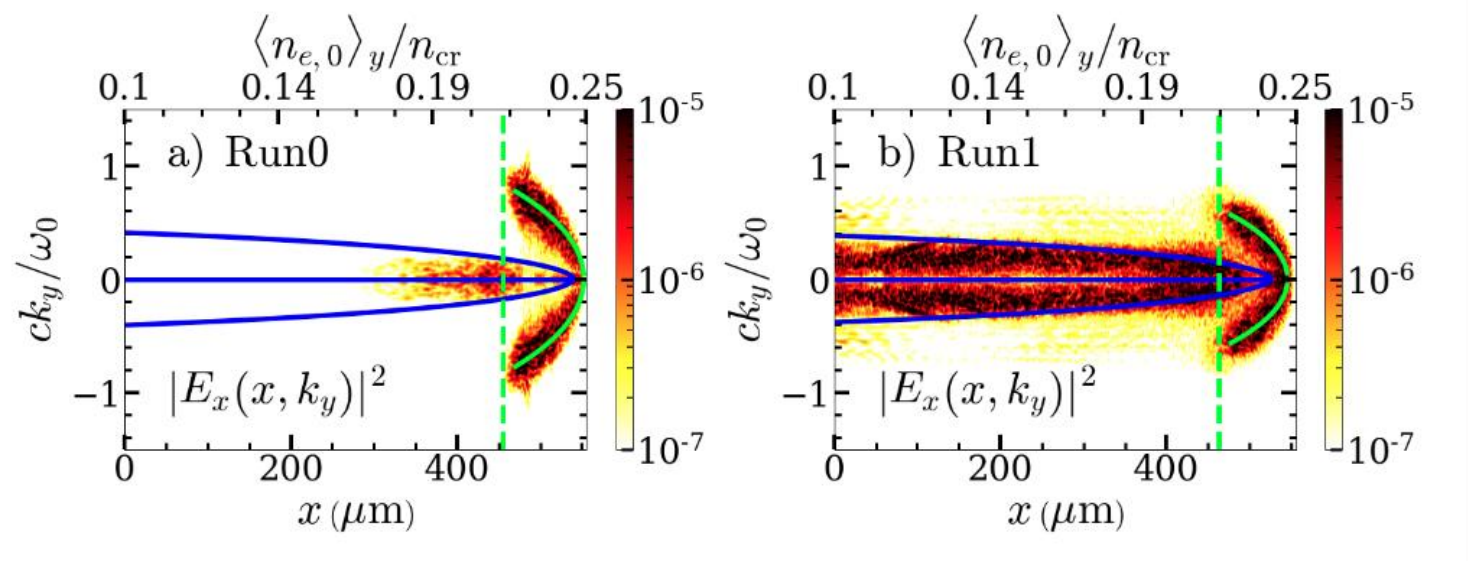

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Overview

01 LPI theory

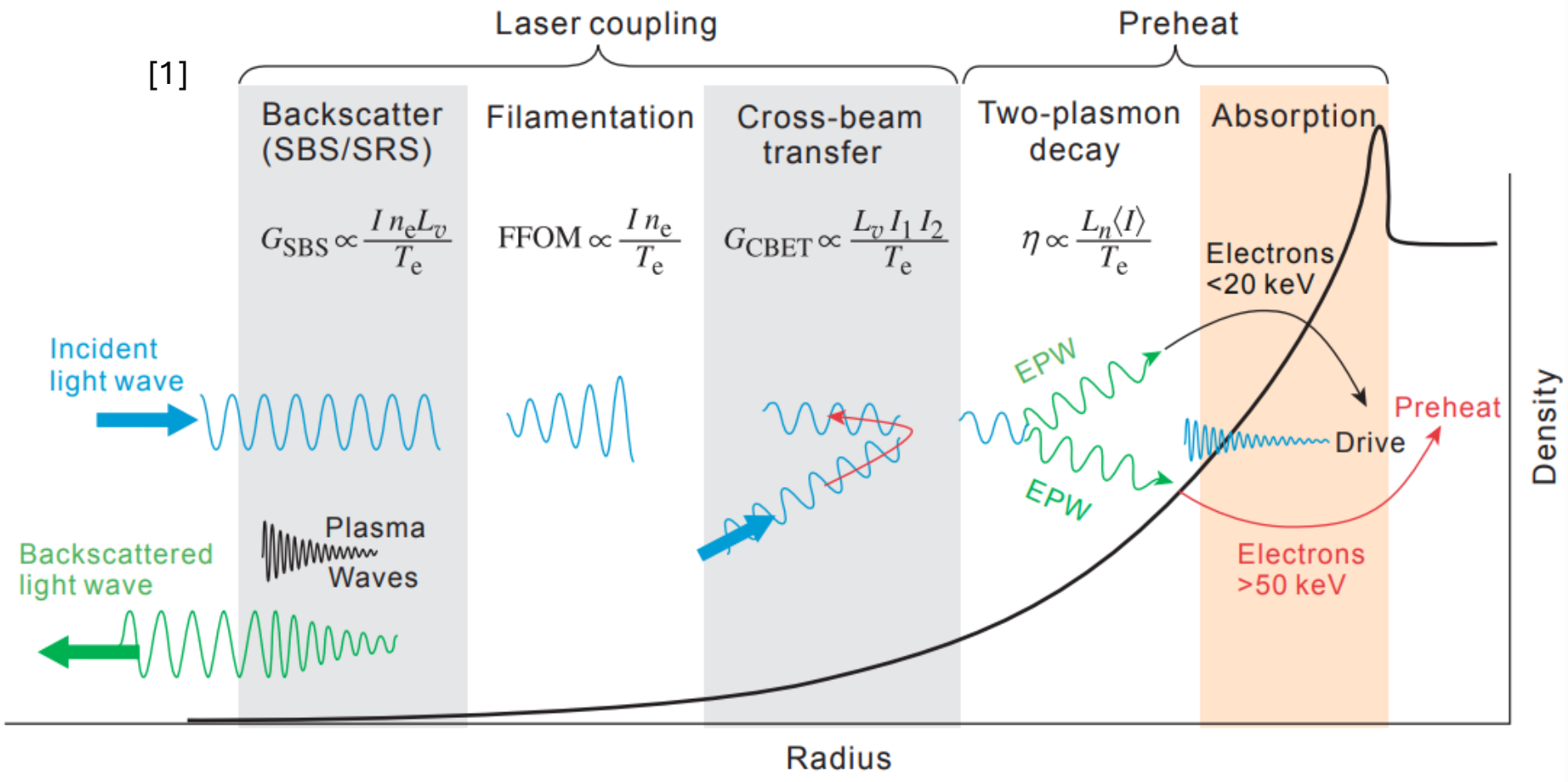
02 Modelling LPI in EPOCH

03 Synthetic diagnostics

04 Experimental comparisons

1. Laser-Plasma Instabilities

[1]



Phase-Matching

$$\mathbf{k}_0 = \mathbf{k}_1 + \mathbf{k}_2$$

$$\omega_0 = \omega_1 + \omega_2$$

Wave-Dispersion

$$\omega_{em}^2 = \omega_{pe}^2 + c^2 k_{em}^2$$

$$\omega_{epw}^2 = \omega_{pe}^2 + 3 v_{th}^2 k_{epw}^2$$

$$\omega_{iaw} = (c_s + u_{flow} \cos\theta)\, k_{iaw}$$

$$\theta \angle (\mathbf{u}_{flow}, \mathbf{k}_{iaw})$$

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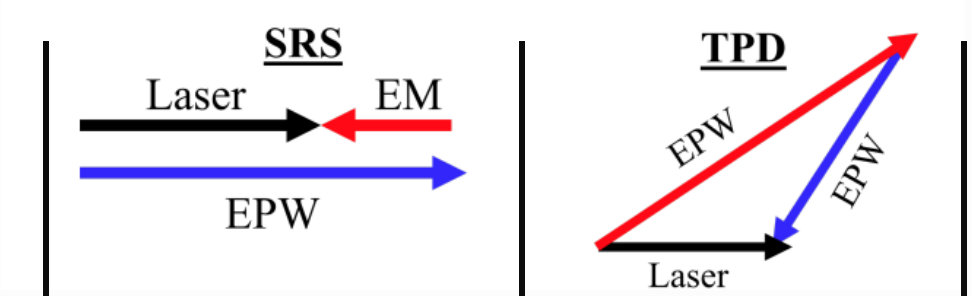
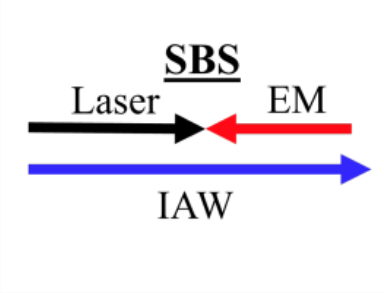
[1] E.M. Campbell et al. *Matter and Radiation at Extremes* 2 (2017) 37-54

1. Key LPI Parameters

		
n_e	$\leq 0.25\, n_{\mathrm{cr}}$	$0.2 - 0.25\, n_{\mathrm{cr}}$	$\leq\ n_{\mathrm{cr}}$
Convective/Absolute	$< 0.25\, n_{\mathrm{cr}}/0.25\, n_{\mathrm{cr}}$	$< 0.25\, n_{\mathrm{cr}}/0.25\, n_{\mathrm{cr}}$	$< n_{\mathrm{cr}}/n_{\mathrm{cr}}$
Convective Gain	$G_{\mathrm{SRS}}\left(n_e, I, \lambda_0, L_n, \Phi\right)$	$G_{\mathrm{TPD}}\left(n_e, I, \lambda_0, L_n, \Phi, T_e\right)$	$G_{\mathrm{SBS}}\left(u, I, \lambda_0, L_u\right)$
LPI Threshold	$I_{\mathrm{SRS}}\left(T_e, L_n, \lambda_0, \Phi\right)$	$I_{\mathrm{TPD}}\left(T_e, L_n, \lambda_0, \Phi\right)$	$I_{\mathrm{SBS}}\left(T_i/T_e, L_u, \lambda_0\right)$

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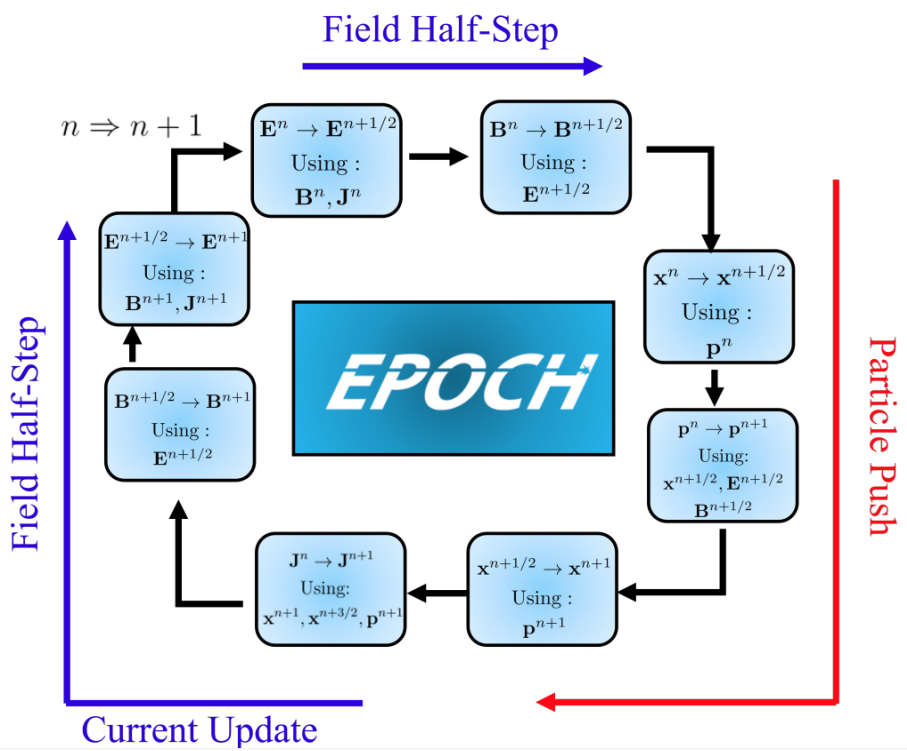
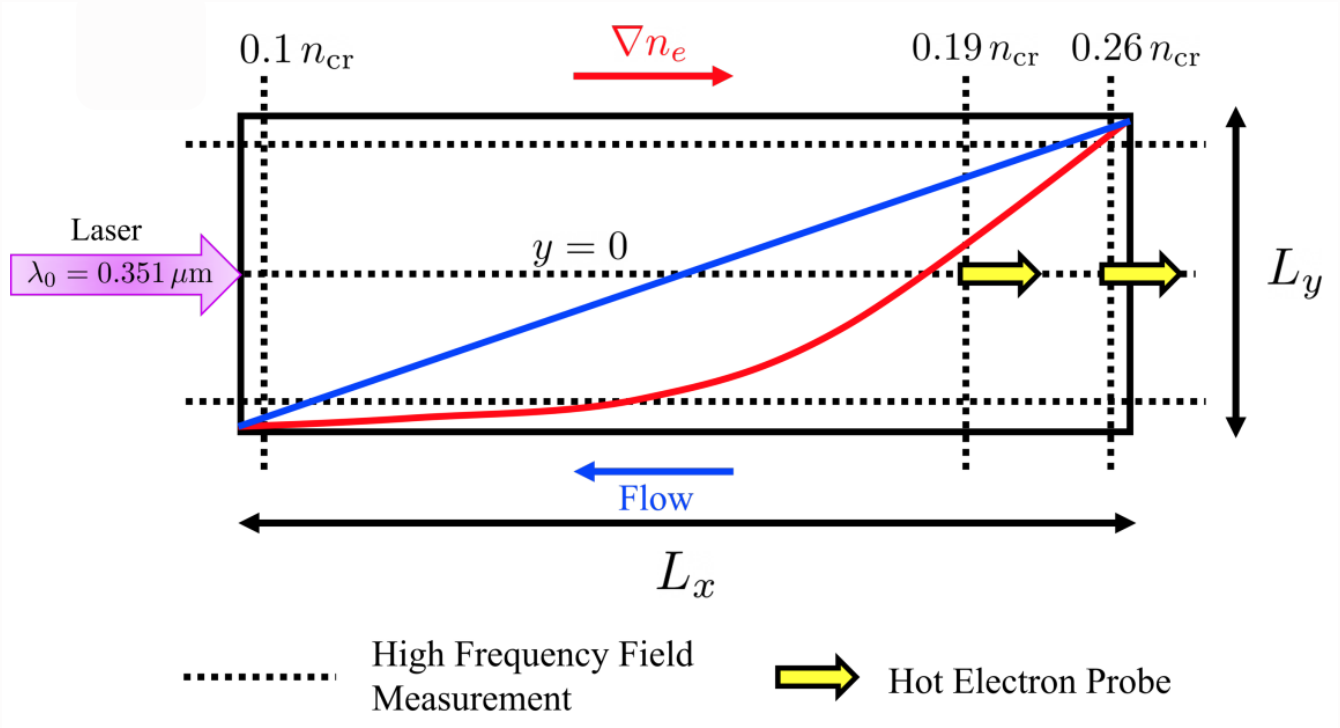
2.Modelling LPI using EPOCH





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2. LPI Simulation Deck – Control and Boundaries

```
begin:control
   # Grid Resolution - 1.2 lambda_D,min
   nx = 38000
   ny = 1200
   t_end = final_time_sims
   # Size of domain
   x_min = 0
   x_max = x_tot_size
   y_min = -y_tot_size
   y_max = y_tot_size
   # Data output flags
   max_accumulator_steps = -1
   restart_snapshot = 1
   allow_missing_restart = T
   use_exact_restart = T
end:control
```

```
begin:boundaries
    bc_x_max_field = open
    bc_x_min_field = simple_laser

    bc_x_max_particle = open
    bc_x_min_particle = open

    bc_y_max_field = periodic
    bc_y_min_field = periodic

    bc_y_max_particle = periodic
    bc_y_min_particle = periodic
end:boundaries
```

- To limit self-heating grid cell number should be N > L / (Debye Length)
- In 2D a useful approach is N = L/(1.2 min(Debye Length)) using third order spline functions

```
make -j COMPILER=gfortran DEFINE=-DPARTICLE_SHAPE_BSPLINE3
```

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2. LPI Simulation Deck – Species (Electron)

```
begin:species
   name = Electron
   charge = -1.0
   mass = 1
   nparticles_per_cell = npart_per_cell
   temperature_ev = temperature_in_ev
   density = if(x lt 0.0, n_min, n_min * exp(x / l_n))
   dumpmask = always

   # For flow, linearly interpolate speed between x_min and x_max
   # Note: x_max is cell-centred, some particles have x beyond this - clamp speed
   drift_px = -mass_e * if(x lt x_max, ((vx_026 - vx_01) * x / x_max + vx_01), vx_026)
end:species
```

```
begin:injector
   boundary = x_max
   species = Electron
   number_density = n_max
   temp_ev = temperature_in_ev
   drift_x = -mass_e * vx_026
   npart_per_cell = npart_per_cell
end:injector
```

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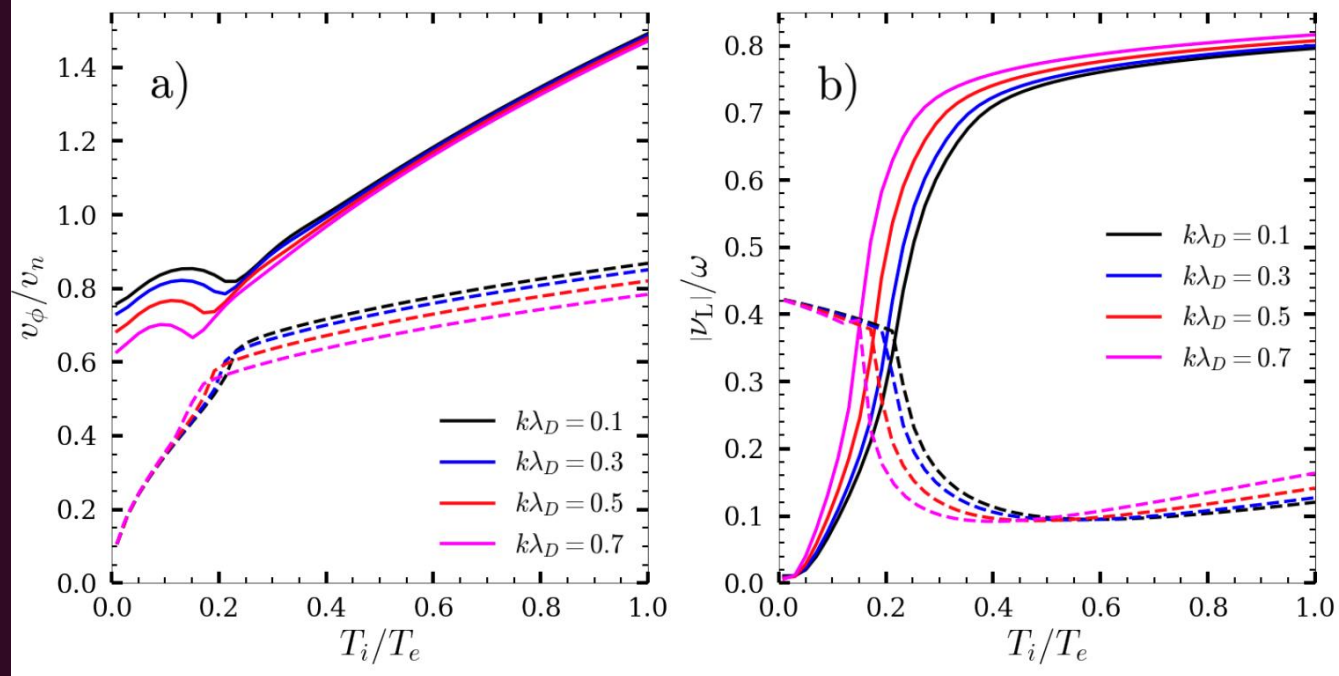
2. LPI Simulation Deck – Species (Ion)

```
# ION PART: Carbon

begin:species
   name = carbon
   charge = 6
   mass = 12 * mp_me
   nparticles_per_cell = npart_per_cell / 2
   number_density = number_density(Electron) / 7
   temperature_ev = ion_temp_in_ev
   dumpmask = always
   immobile = F

   # For flow, linearly interpolate speed between x_min and x_max
   # Note: x_max is cell-centred, some particles have x beyond this - clamp speed
   drift_px = -12 * mass_p * if(x lt x_max, ((vx_026 - vx_01) * x / x_max + vx_01), vx_026)
end:species


# ION PART: proton

begin:species
   name = proton
   charge = 1
   mass = mp_me
   nparticles_per_cell = npart_per_cell / 2
   number_density = number_density(Electron) / 7
   temperature_ev = ion_temp_in_ev
   dumpmask = always
   immobile = F

   # For flow, linearly interpolate speed between x_min and x_max
   # Note: x_max is cell-centred, some particles have x beyond this - clamp speed
   drift_px = -mass_p * if(x lt x_max, ((vx_026 - vx_01) * x / x_max + vx_01), vx_026)
end:species
```



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2. LPI Simulation Deck – Collisions

```
begin:collisions
    use_collisions = T
    use_nanbu = T

    # Speed-up flags
    use_cold_correction = F     # Assume e- over 10 eV
    rel_cutoff = 0.01           # Use lab-frame if relativistic p change under 1%
    coll_n_step = 10            # Collide once every 10 steps, with 10x the scatter
    back_update_dt = 100.0e-15 # Recalculate background ni every 100 fs (not every step)
    coulomb_log = 7.2           # Assume fixed Coulomb logarithm

    # Choose colliding species
    collide = none
    collide = Electron proton background
    collide = Electron carbon background
    collide = Electron Electron on
end:collisions
```

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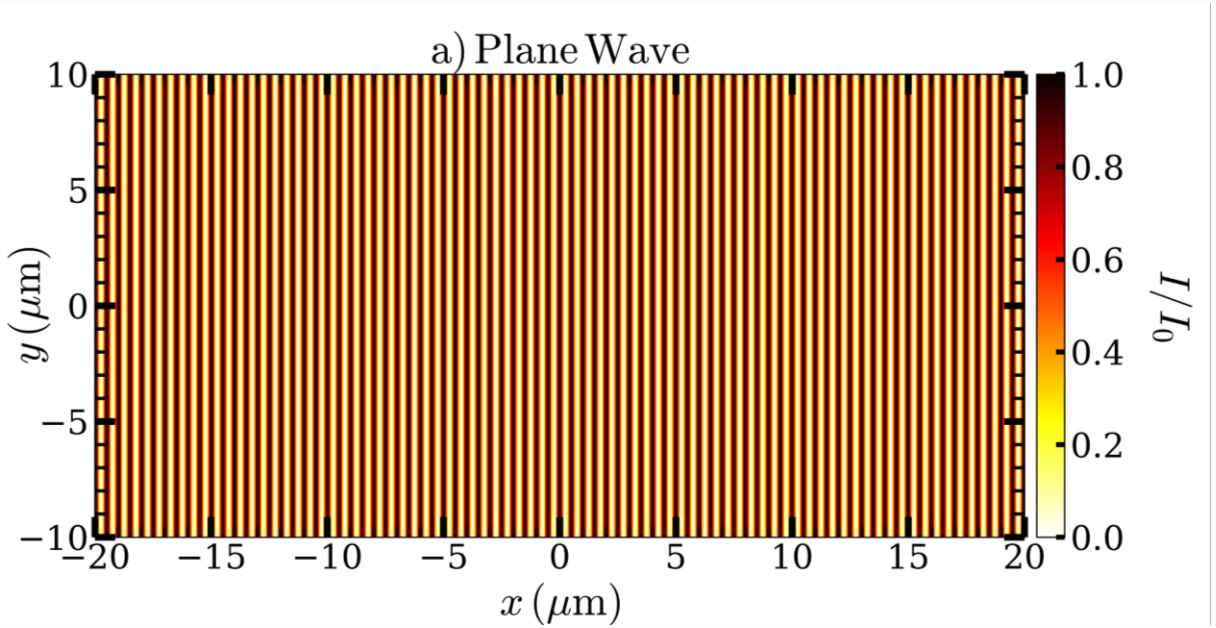
2. LPI Simulation Deck – Lasers (Plane Wave)

```
begin:laser
   boundary = x_min
   t_profile = semigauss(time,1.0,0.001,9.892473060642255e-14)
   t_end = final_time_laser
   intensity_w_cm2 = 1.0e16
   omega = omega0
   polarisation = 45
   profile = 1
end:laser
```



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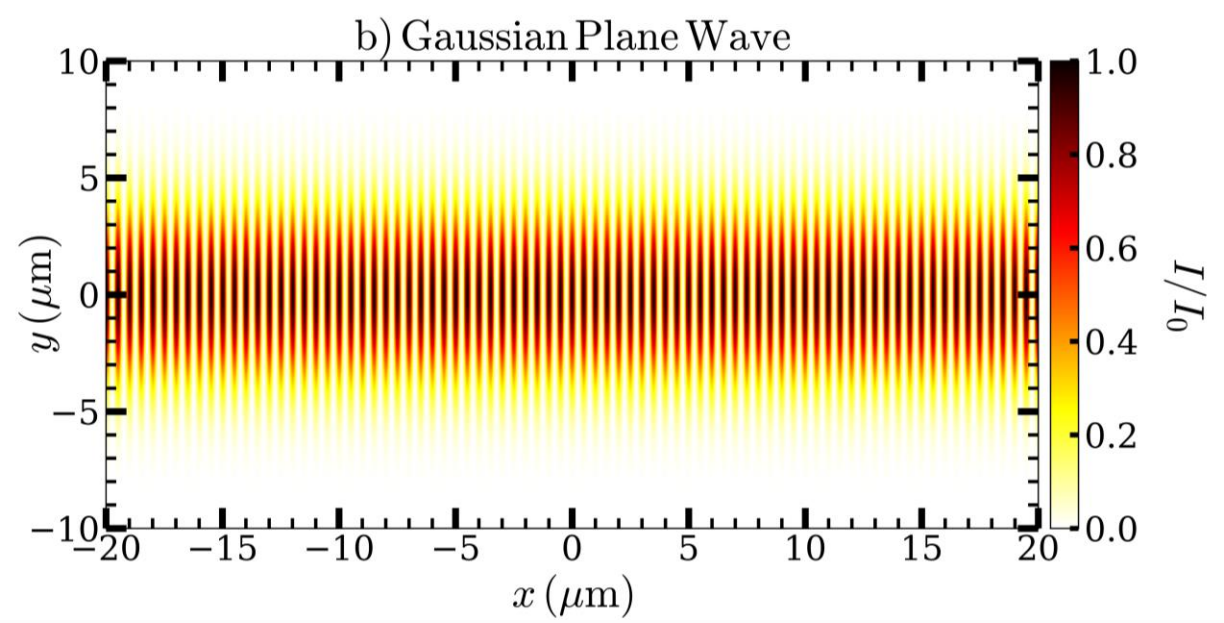
2. LPI Simulation Deck – Lasers (Gaussian)

```
begin:laser
   boundary = x_min
   t_profile = semigauss(time, 1, 0.001, rise)
   t_end = final_time_laser
   intensity_w_cm2 = peak_intensity
   omega = omega0
   polarisation = 0
   profile = gauss(y, 0, w_0)
end:laser
```

$$\mathbf{E}(x,y) = E_0 \exp\left(-4\ln 2 \frac{(y-y_0)^2}{w^2}\right) \exp(ikx)\hat{\mathbf{p}}$$



$$r_{\text{Speckle}} \sim F_{\#}\lambda_0$$

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3. LPI Simulation Deck – Lasers (Focused Gaussian)

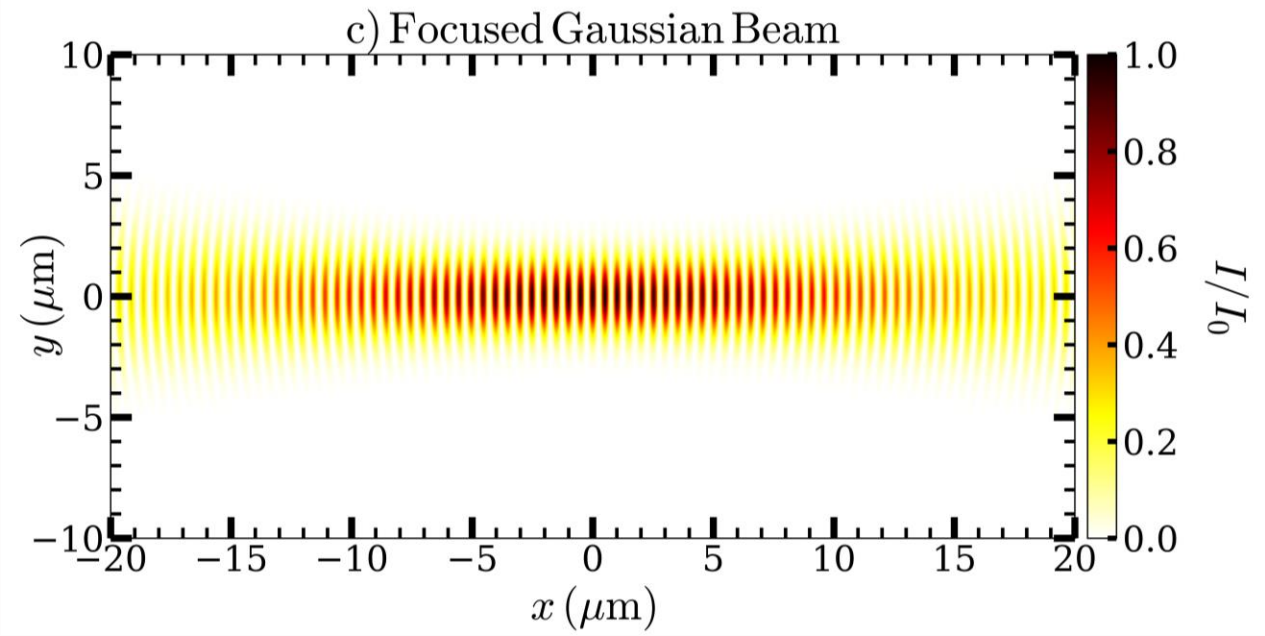
```
# SPECKLE PARAMETERS

   # Force a Gaussian focus with focal spot halfway along x domain
   foc_dist = l_tot * 0.5                                # Distance to focus
   las_k = 2.0 * pi / lambda0                            # Laser vacuum wavenum.
   w0 = I_fwhm / sqrt(2.0 * loge(2.0))                   # Beam Waist
   ray_rang = pi * w0^2 / lambda0                        # Rayleigh range
   w_boundary = w0 * sqrt(1.0 + (foc_dist/ray_rang)^2)   # Waist on boundary
   I_boundary = I_peak_Wcm2 * (w0 / w_boundary)^2        # Intens. on boundary
   rad_curve = foc_dist * (1.0 + (ray_rang/foc_dist)^2)  # Boundary curv. rad.
   gouy = atan(-foc_dist/rad_curve)                      # Phase correction

end:constant
```

```
begin:laser
   boundary = x_min
   t_profile = semigauss(time,1.0,0.001,9.892473060642255e-14)
   t_end = final_time_laser
   intensity_w_cm2 = I_boundary
   omega = omega0
   polarisation = 0
   # Speckle has a focussing Gaussian profile
   phase = las_k * y^2 / (2.0 * rad_curve) - gouy
   profile = gauss(y, 0, w_boundary)
end:laser
```

$$\mathbf{E}(x,y) = E_0 \frac{w_0}{w(\xi)} \exp\left(-\frac{(y-y_0)^2}{w^2(\xi)}\right) \exp\left(-ikx - ik\frac{y^2}{2R(\xi)} - i\psi(\xi)\right) \hat{\mathbf{p}}$$



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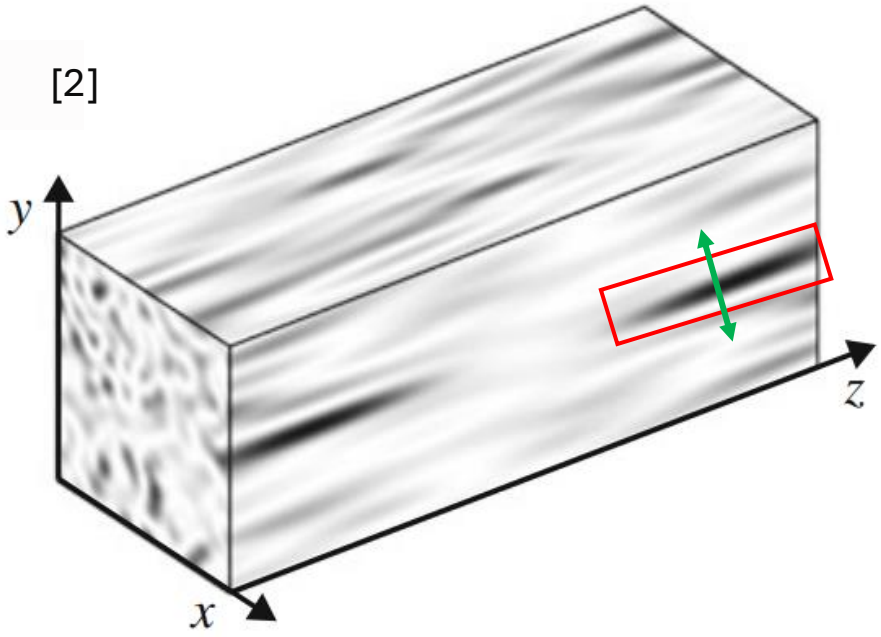
2. Speckle Patterns

[2]



- Experiments often use **Random Phase Plates (RPP)** and **Smoothing by Spectral Dispersion (SSD)** to help limit LPI

Wavefront Curvature terms (similar to focussed Gaussian)

$$\mathbf{E}(x_b, y, t) \propto \exp[i\phi(x_b, y, t)] \sum_{n=1}^{N_y} A_n(t)\,\hat{\mathbf{p}}(t) \exp[i\Psi_n(x_b) + i\Phi_n(t)]$$

$$\Psi(x_b) = \varphi_n + \frac{k_0 x_b n^2}{8F_\#}$$

$$\Phi_n(t) = \sum_i \beta_i \sin[2\pi\nu_{m,i}(t - nt_D)]$$

Static phase (RPP) - First term is the applied random phase, second is optics specific (F-number of lens)

Time varying phase (SSD) - Motion of speckles controlled by applied frequency/bandwidth

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[2] P. Michel *Introduction to Laser-Plasma Interactions* 2023 (Springer, ISBN: 978-3-031-23424-8)

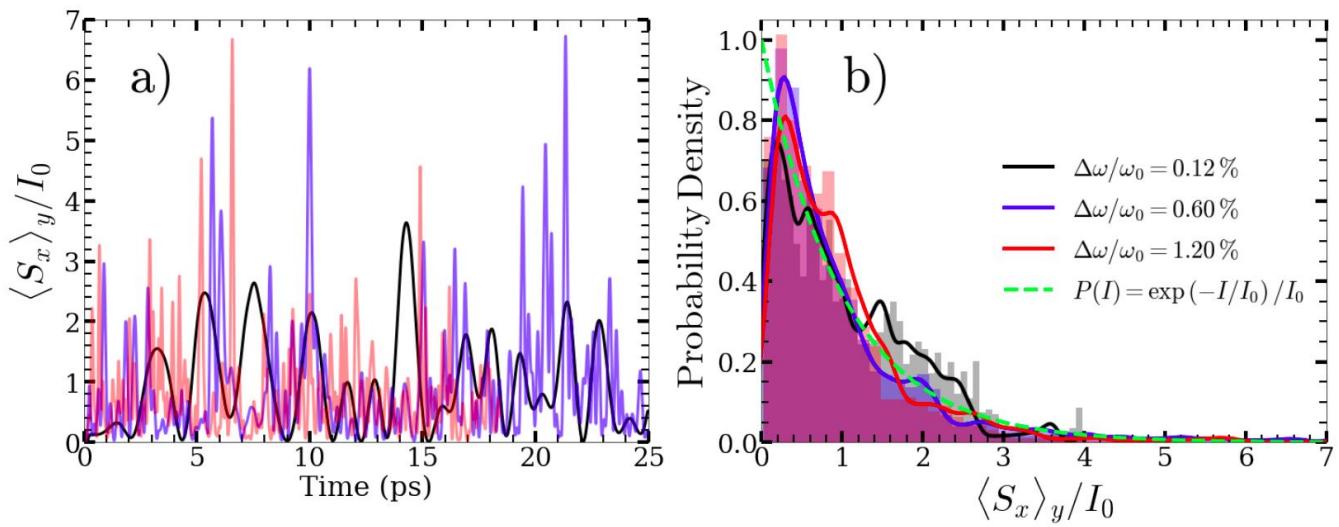
2. LPI Simulation Deck – Lasers

```
begin:laser
   boundary = x_min
   t_profile = semigauss(time,1.0,0.001,9.892473060642255e-14)
   t_end = final_time_laser
   amp = 19908008248.262657
   omega = 0.9942550550157717 * omega0
   polarisation = 0
   profile = 1
   phase = 2.538202250911291
end:laser

begin:laser
   boundary = x_min
   t_profile = semigauss(time,1.0,0.001,9.892473060642255e-14)
   t_end = final_time_laser
   amp = 19908008248.262657
   omega = 0.9942724887059953 * omega0
   polarisation = 0
   profile = 1
   phase = 2.82744819118835
end:laser

begin:laser
   boundary = x_min
   t_profile = semigauss(time,1.0,0.001,9.892473060642255e-14)
   t_end = final_time_laser
   amp = 19908008248.262657
   omega = 0.9943691103534318 * omega0
   polarisation = 0
   profile = 1
   phase = 5.829338106299704
end:laser
```

$$\mathbf{E} = \hat{\mathbf{y}} \sum_{n=1}^{N} A_n(\omega_n) \exp[i\,(\mathbf{k}(\omega_n)\cdot\mathbf{r} - \omega_n t + \varphi_n)]$$

$$\omega_n \sim \mathcal{U}\left(\omega_0 - \frac{\Delta\omega}{2}, \omega_0 + \frac{\Delta\omega}{2}\right) \qquad \varphi_n \sim \mathcal{U}(-\pi, \pi)$$

$$A_n = \sqrt{\frac{2\Delta\omega I(\delta\omega_n)}{N c \epsilon_0 \eta(\omega_n)}}$$



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3. Frequency Filtered Diagnostics

- To extract specific LPI behaviour we need to filter the outputted field components

- Set-up bandpass filter

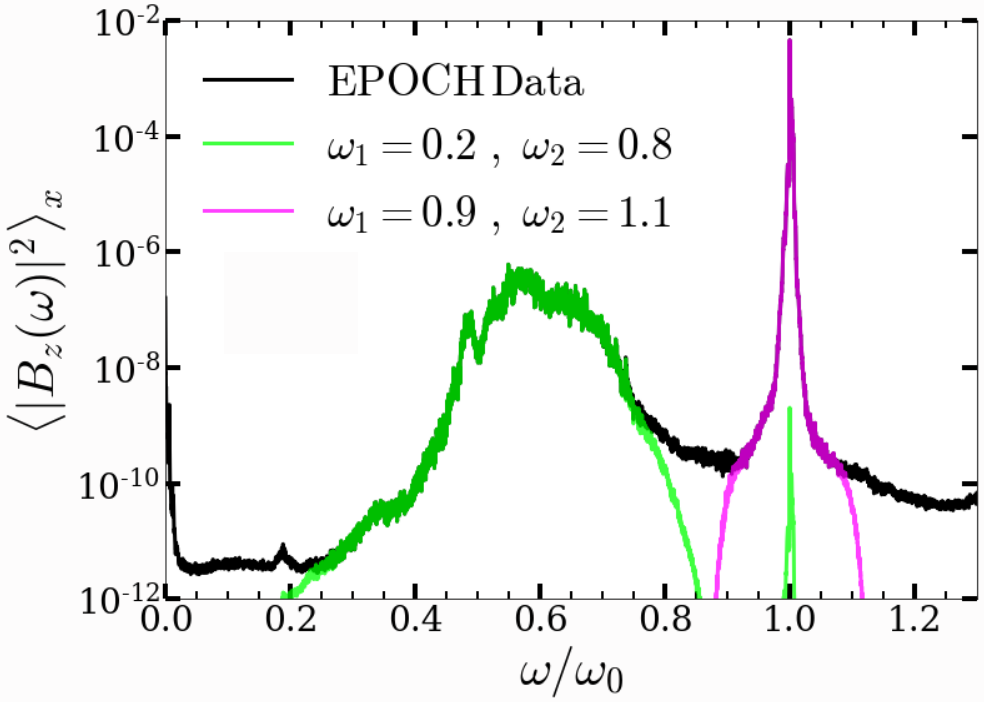
$$h_{\mathrm{BP}}[n] = \frac{\sin\left(\bar{\omega}_2\left(n-\frac{M}{2}\right)\right)}{\pi\left(n-\frac{M}{2}\right)} - \frac{\sin\left(\bar{\omega}_1\left(n-\frac{M}{2}\right)\right)}{\pi\left(n-\frac{M}{2}\right)}, \quad n = 0, 1, \ldots, M$$

$$w_{\mathrm{Hanning}}[n] = 0.5\left(1 - \cos\left(\frac{2\pi n}{M}\right)\right)$$

- Extracted filtered field signal (For longer signals FFT preferred over convolution)

$$F_{\mathrm{filtered}} = \mathrm{IFFT}\big(\mathrm{FFT}(F)\cdot\mathrm{FFT}(h)\big)$$

- Can be done using scipy.fftconvolve



- 0.2-0.8 omega – **SRS light**
- 0.9-1.1 - **Laser/SBS light**

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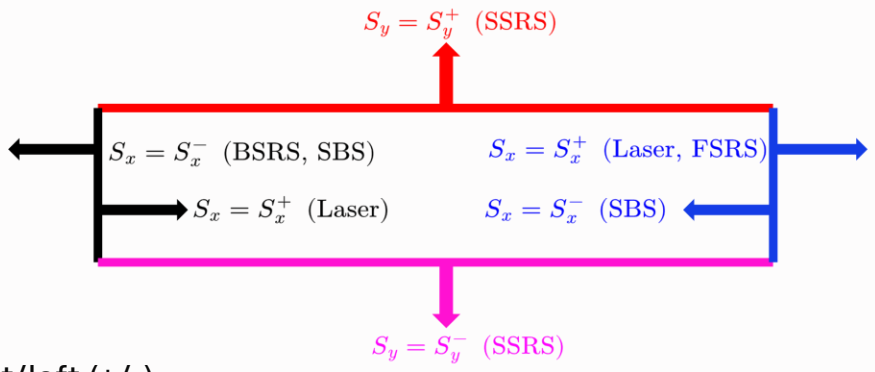
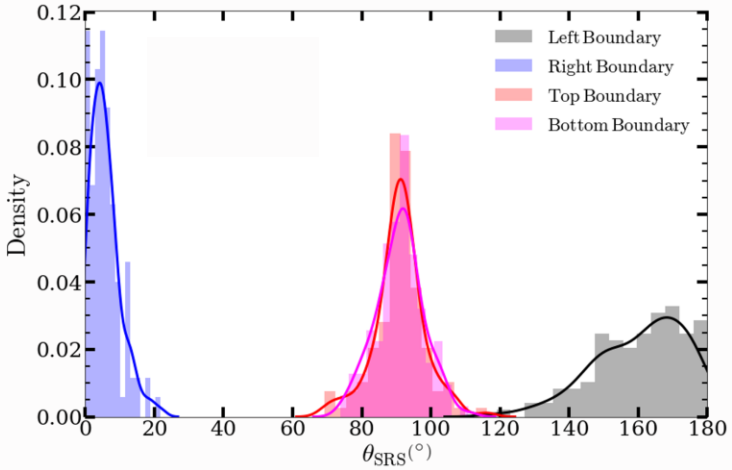
3. Directional Poynting flux

- It is useful to extract directional information to discern wave modes from one another
- e.g Laser and SBS light have same frequency, but travel in opposite directions





$$S_x^{\pm} = \frac{\epsilon_0 c}{4}\left[(E_y \pm cB_z)^2 + (E_z \mp cB_y)^2\right]$$

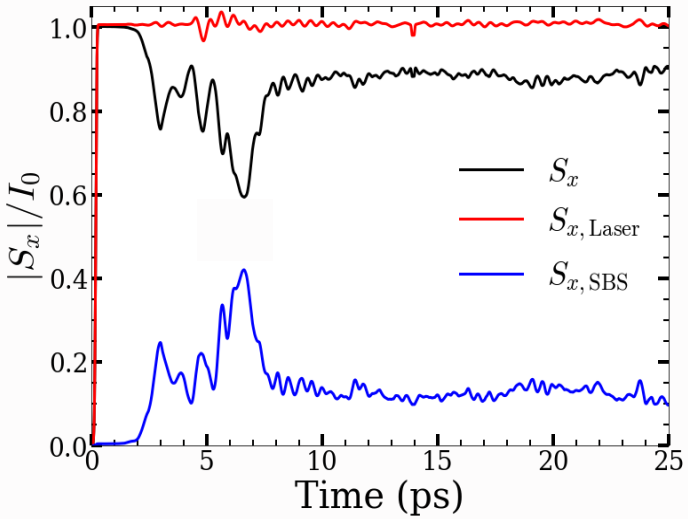
- Right/left (+/-)

$$S_y^{\pm} = \frac{\epsilon_0 c}{4}\left[(E_z \pm cB_x)^2 + (E_x \mp cB_z)^2\right]$$

- Upward/downward (+/-)

- Reflectivity/Transmission from filtered flux at given boundary

$$R\,,\,T = \frac{\langle S \rangle}{\mathcal{P}_0}$$



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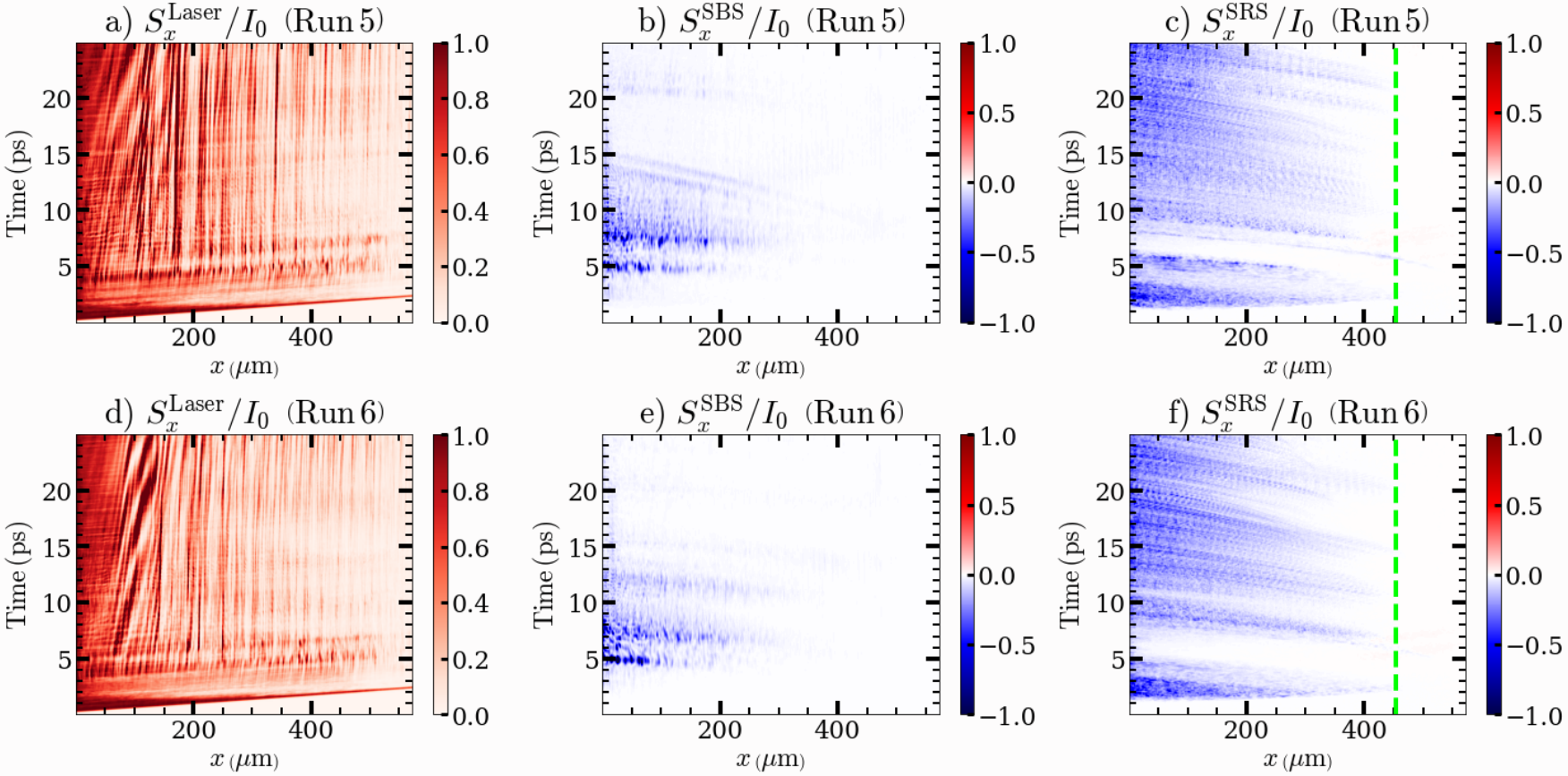
3. Directional Poynting flux



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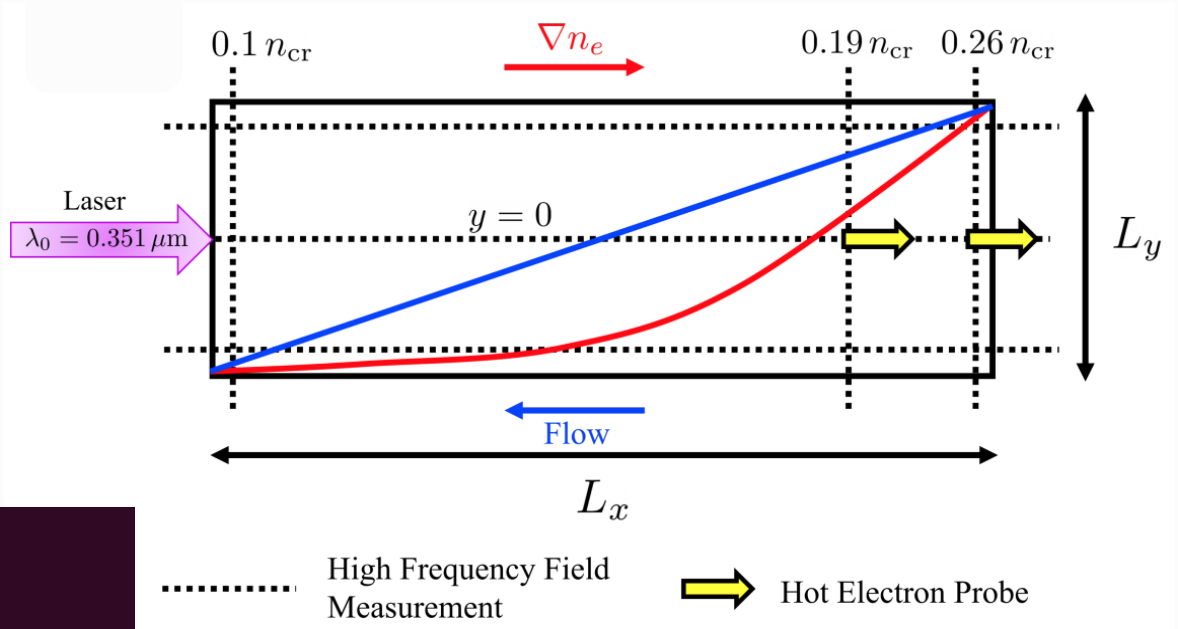
3. LPI Simulation Deck – High Frequency Data

```
# RIGHT BOUNDARY

begin:subset
   name = x_right
   x_min = x_max - 6 * dx
   x_max = x_max - 5 * dx
end:subset


# LOWER BOUNDARY

begin:subset
   name = y_lower
   y_min = y_min + 5 * dy
   y_max = y_min + 6 * dy
end:subset
```




```
begin:output
   file_prefix = acc_field
   name = acc_field
   dt_snapshot = 100 * femto
   ex = x_nc19 + strip_y0 + x_left + x_right + y_lower + y_upper + accumulate + single
   ey = x_nc19 + strip_y0 + x_left + x_right + y_lower + y_upper + accumulate + single
   bz = x_nc19 + strip_y0 + x_left + x_right + y_lower + y_upper + accumulate + single
   dt_accumulate = 0.05 * femto
   dump_first = F
end:output
```

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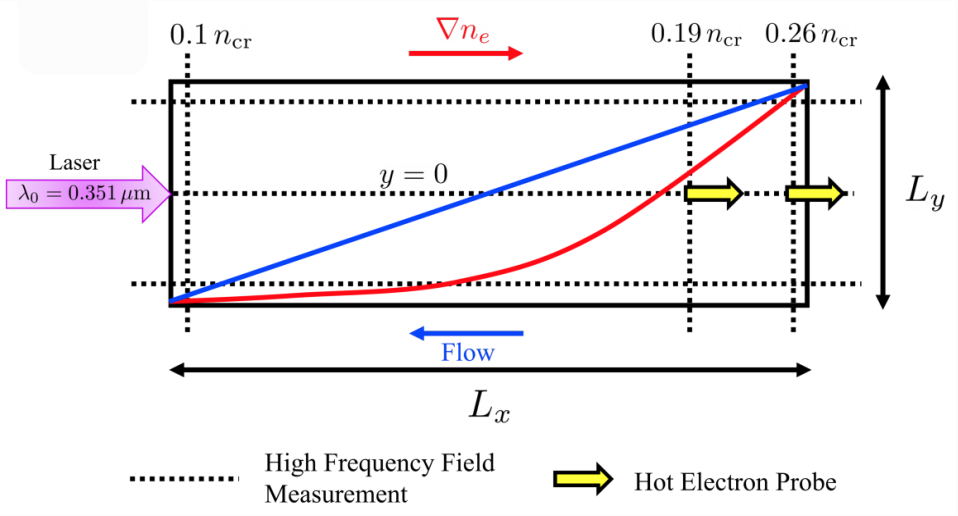
3. Hot-electron Diagnostics - Temperature

- Extract outgoing electron flux distribution from probes

$$F_e(E,y,t) \quad \Rightarrow \quad f_e(E) = \frac{1}{\Delta y \Delta t}\langle F_e(E,y,t)\rangle_{y,t}$$

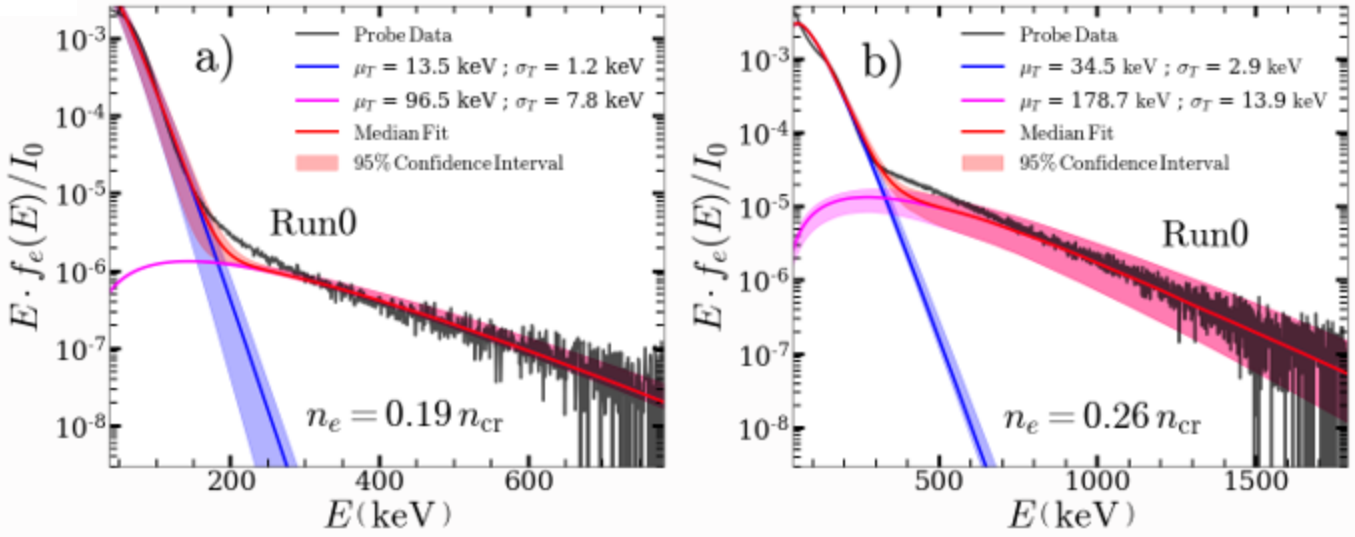


- Fit electron flux distribution (Brute force, Markov – Chain Monte-Carlo)

$$F_{\text{Model}}(E,\theta) = 2\sqrt{\frac{E}{\pi}} \cdot \sum_j A_j \left(\frac{1}{k_B T_j}\right)^{3/2} \exp\left(-\frac{E}{k_B T_j}\right)$$

- Estimate single hot-electron temperature

$$T_{\text{Hot}} = \frac{\sum_j \langle T_j \rangle \cdot \bar{F}_j}{\sum_j \bar{F}_j}$$



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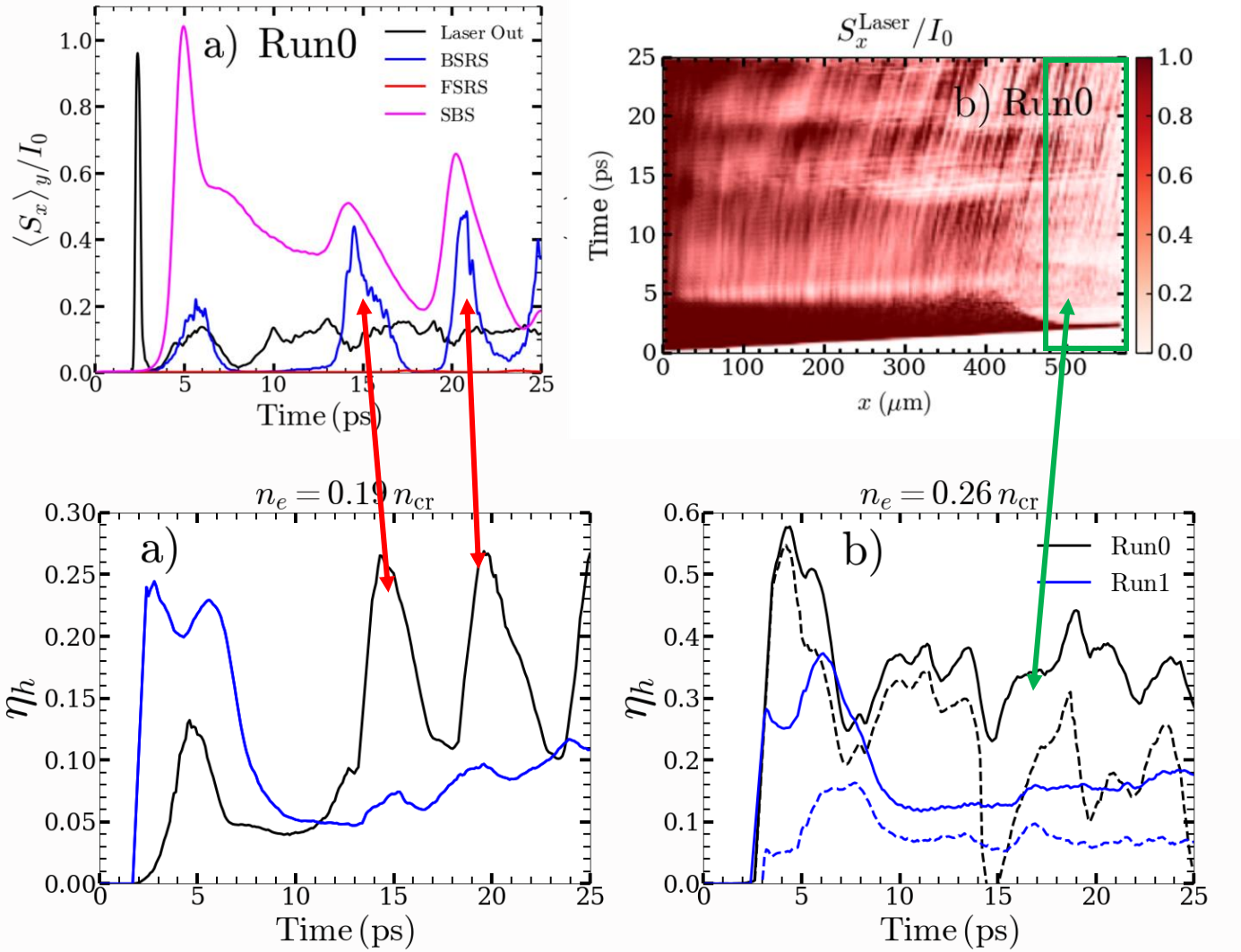
3. Hot-electron Diagnostics – Conversion efficiency

- From binned probe data we can estimate the time-resolved hot-electron flux via:

$$\eta_{\mathrm{Hot}}(t) = \frac{1}{\Delta y \Delta t I_0} \sum_{E>\epsilon} \langle F_e(\epsilon, y, t) \rangle_y$$

- Flux diagnostic used along with filtered Poynting flux data to deduce LPI presence



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3. LPI Simulation Deck - Probes

- name – name of the set up probe
- ek_min/max - minimum and maximum kinetic energy to record for passing species
- include_species – Name the species to record
- normal – Direction to record (+1 = Forward, -1 = Backward)
- point – Location of probe = x (1D), (x,y) (2D)

```
# PROBES

# PROBE FOR OUTGOING ELECTRONS

begin:probe
   name = outgoing_e_probe_
   dumpmask = always
   ek_min = 10 * e_te
   ek_max = -1
   include_species = Electron
   normal = (1.0, 0.0)
   point = (x_max - 15.0e-6, 0)
end:probe


# PROBE FOR ELECTRONS AT 0.19nc

begin:probe
   name = conv_e_probe_
   dumpmask = always
   ek_min = 10 * e_te
   ek_max = -1
   include_species = Electron
   normal = (1.0, 0.0)
   point = (nc_019, 0)
end:probe
```

```
begin:output
   file_prefix = probes
   name = probes
   dt_snapshot = 100 * femto
   particle_probes = always
   dump_first = F
end:output
```

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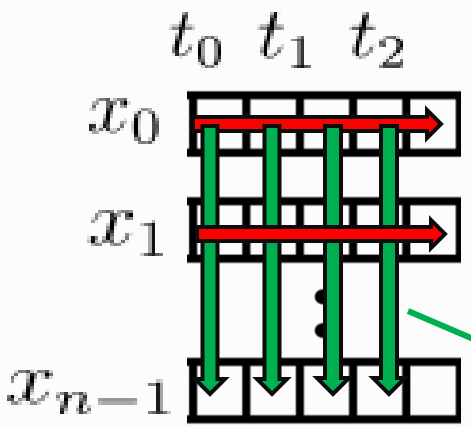
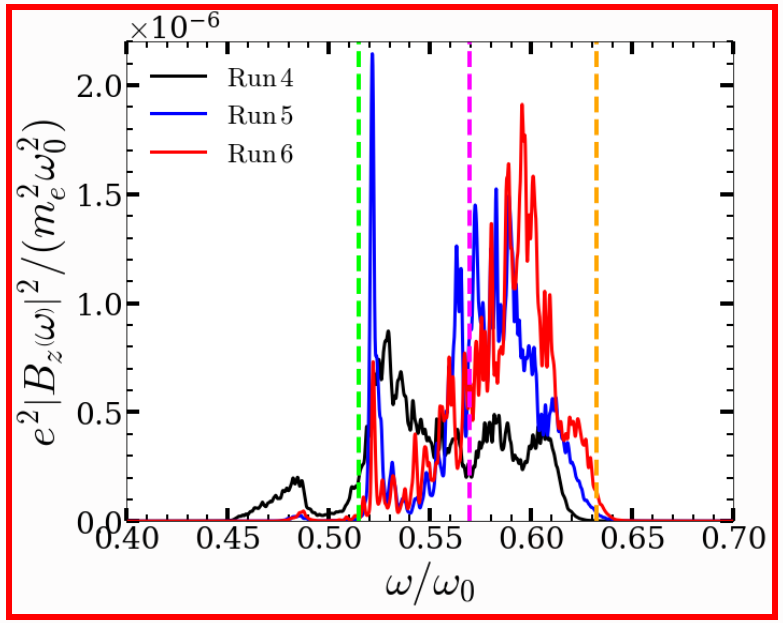
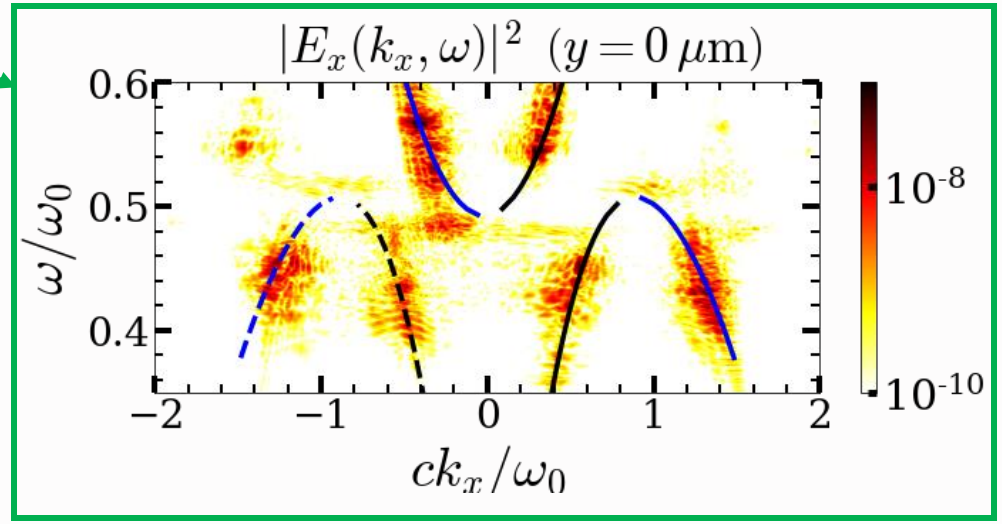
3. Fourier Transform Diagnostics – FFT





- B-field components useful to investigating EM waves (Also Ey and Ez)



- Ex-field useful for investigating EPW waves

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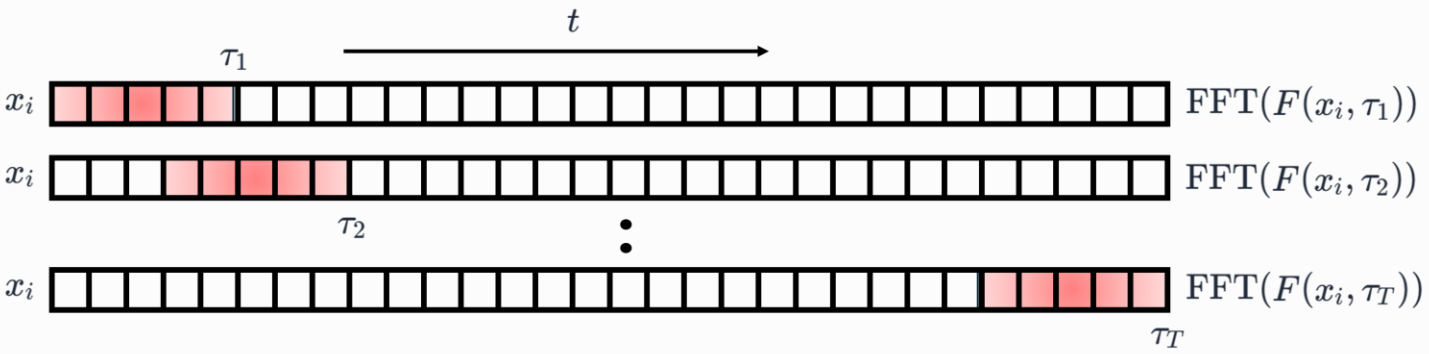
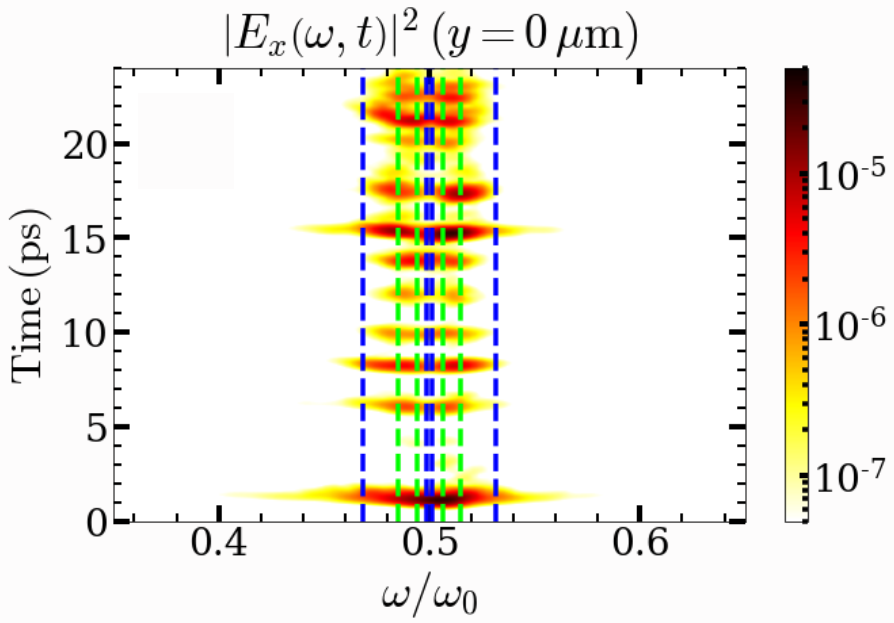
3. Fourier Transform Diagnostics – STFT





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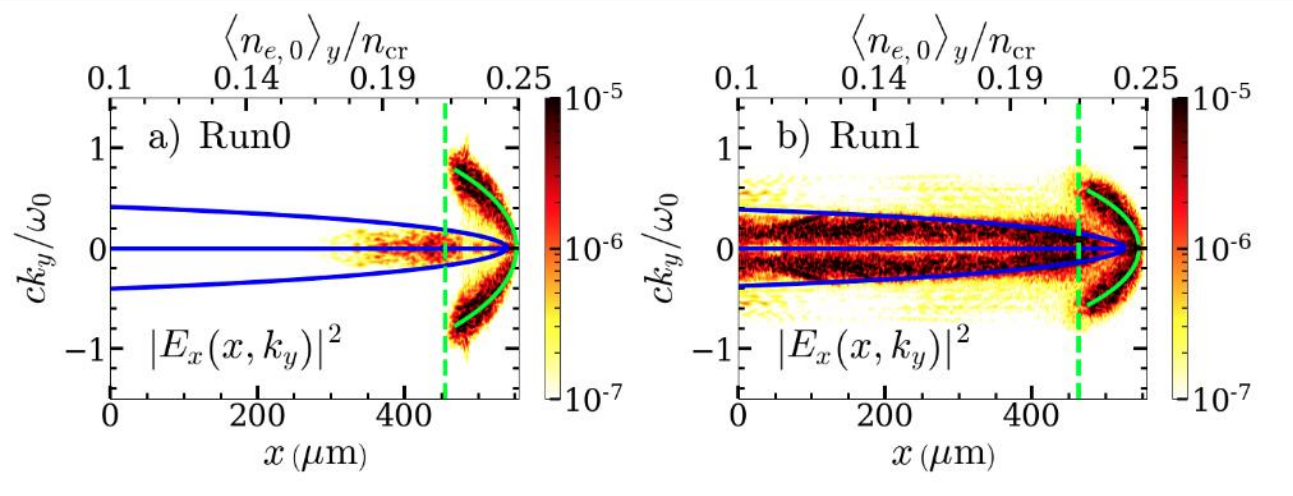
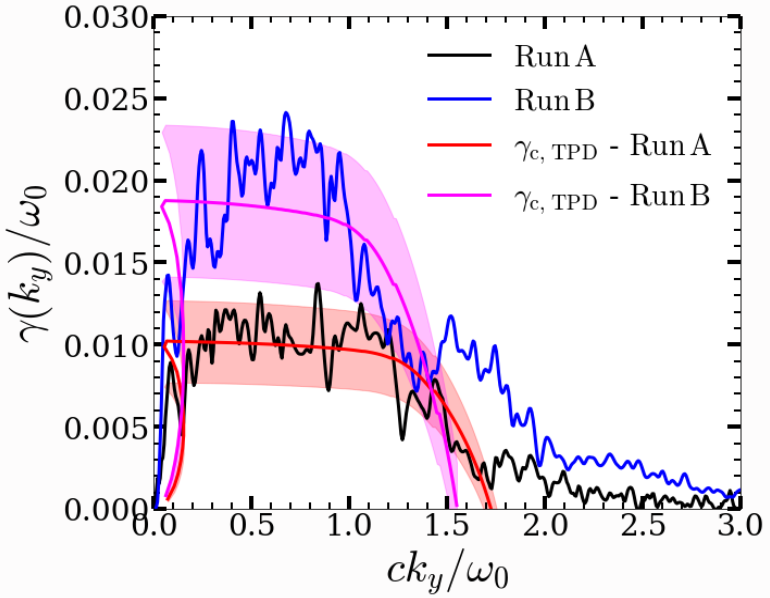
3. LPI Simulation Deck – Bulk outputs

- Useful LPI information can come from bulk 2D grid data
- 2D FFT E_x and n_e/n_i perturbations can show presence of EPW/IAW behaviour
- Transverse FFT along the density gradient can aid in finding where specific LPI are occurring
- Can then use to extract growth rates and asses convective/absolute behaviour

$$\delta n_e, E_x \propto e^{\gamma t} \quad \Longrightarrow \quad \gamma\left(k_y\right) = \left\langle \frac{d}{dt} \ln\left[\delta n_e\left(k_y, t\right)\right] \right\rangle_t$$





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4. Comparisons to Experiments - PALS

[3]

	Simulation	R_{BSRS} (%)	R_{SBS} (%)	T_{Hot} (keV)		η_h (%) (≤ 100 keV)		η_h (%) (> 100 keV)	
				$0.26\ n_{cr}$	$1.05\ n_{cr}$	$0.26\ n_{cr}$	$1.05\ n_{cr}$	$0.26\ n_{cr}$	$1.05\ n_{cr}$
Low Intensity	Run A	0.0	47.0	12 ± 2	17 ± 3	1.7	2.4	0.1	0.2
High Intensity	Run B	1.7	37.0	39 ± 6	37 ± 5	1.8	1.3	1.0	0.7

[4] **Experiment -** SRS reflectivity ~ **2 %**, SBS reflectivity~ **20-30 %**
Hot electron temperature ~ **20-40 keV**, Hot electron conversion ~ **1-2%**

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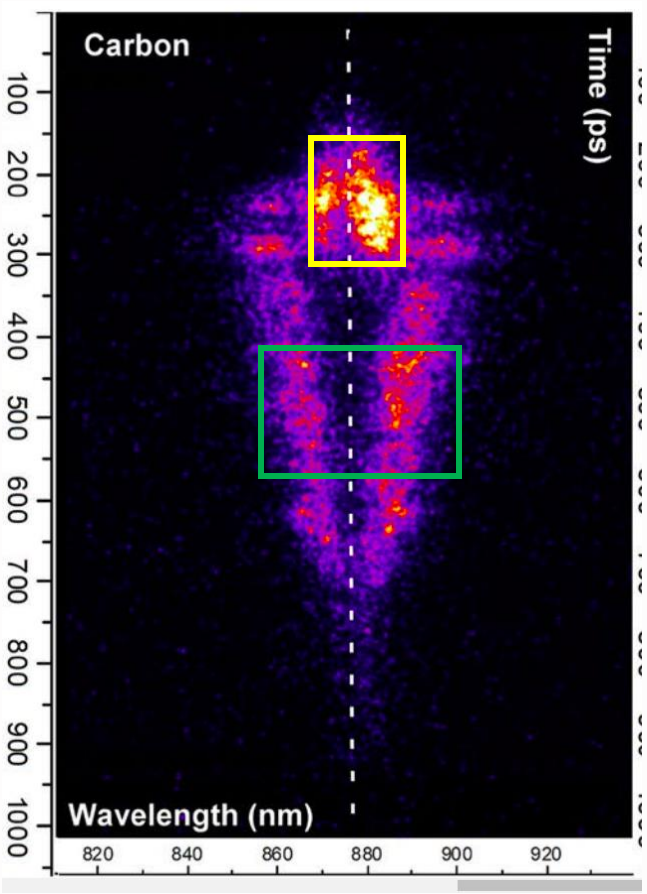
[3] B. Gosling et al. *Phys. Plasmas* 32, 092705 (2025)

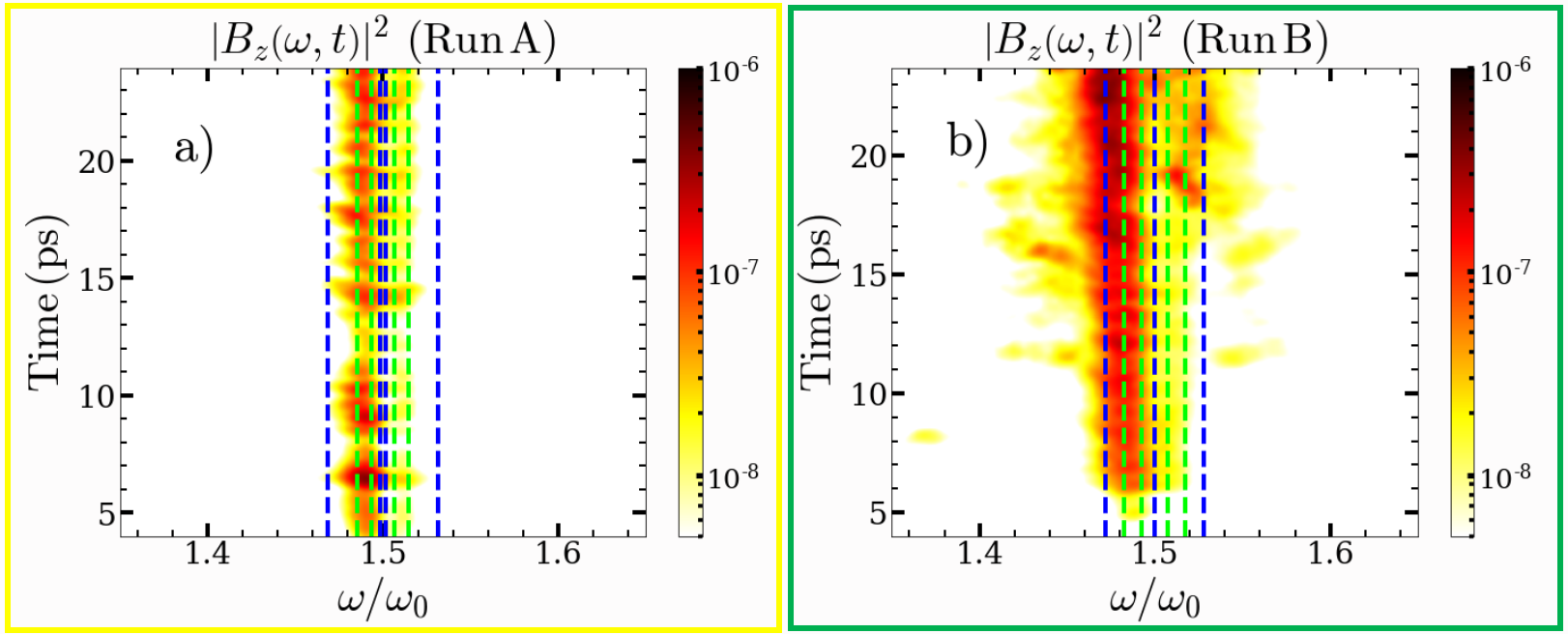
[4] G., Cristoforetti et al. *Scientific Reports* 13, 20681 (2023)

4. Comparisons to Experiments - PALS

[4] **PALS**



[3] **EPOCH**



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[3] B. Gosling et al. *Phys. Plasmas* 32, 092705 (2025)

[4] G., Cristoforetti et al. *Scientific Reports* 13, 20681 (2023)

Git Repository with EPOCH LPI Python scripts:

https://github.com/epochpic/Python-LPI

Email if you want access (B.G.G.J.Gosling@warwick.ac.uk)

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